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Generating Synthetic Interviews in a Psychiatric Setting using a Double Model Approach

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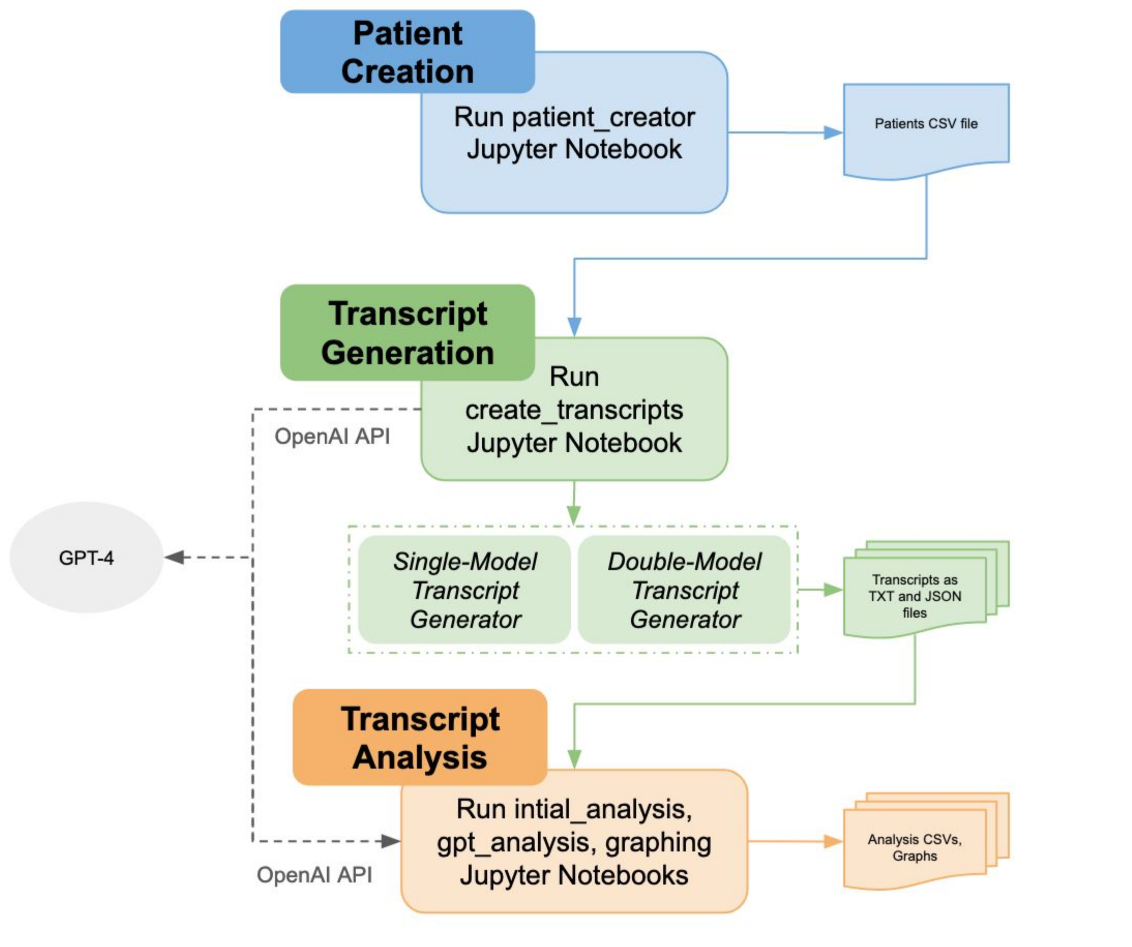

Figure 1. Operational Steps for Synthetic Interview Generation

Synthetic Interview Transcript creation is grouped into three main steps with associated Jupyter Notebook files.

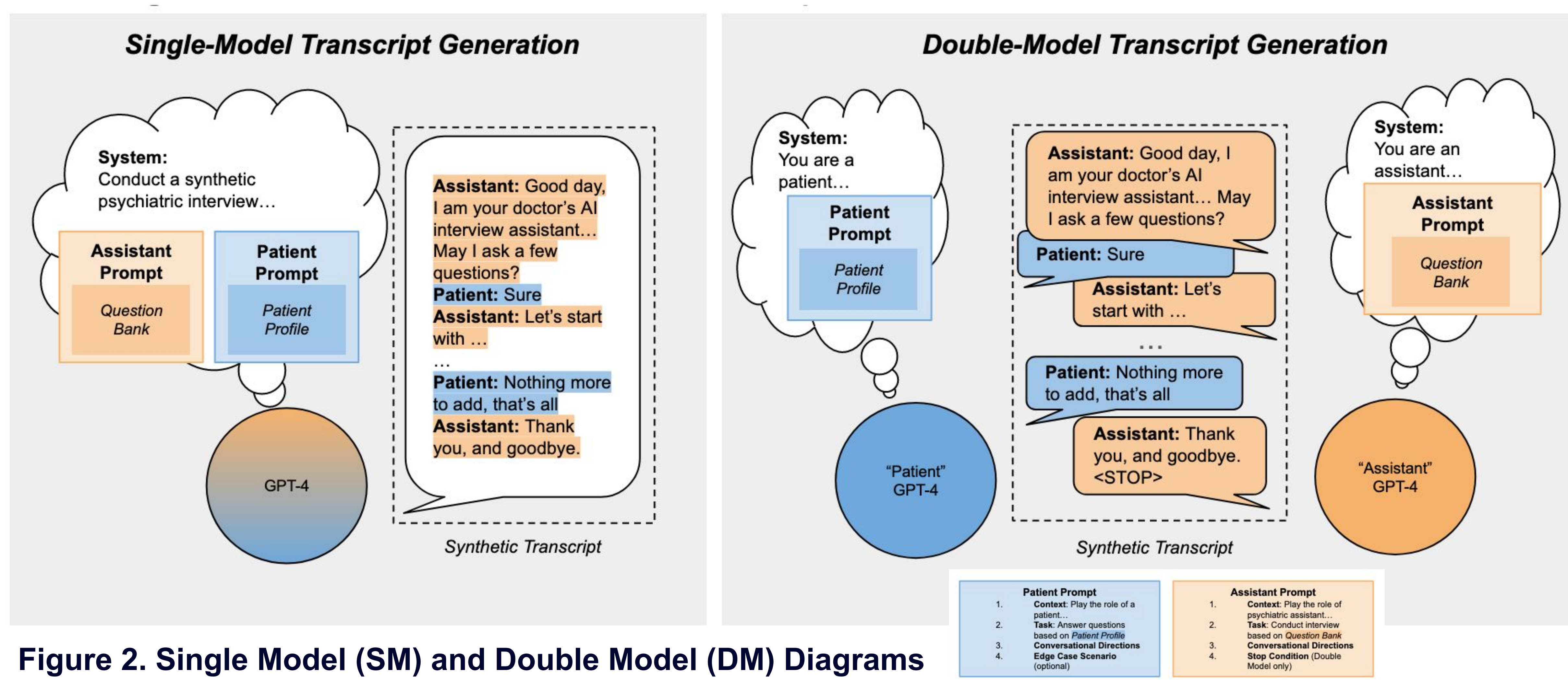

Figure 2. Single Model (SM) and Double Model (DM) Diagrams

In the Single Model approach, one GPT-4 model creates a transcript based on a question bank and the patient profile. In the Double Model approach, two GPT-4 models create the transcript by interacting with each other. The "Assistant" model is instructed with the question bank, and the "Patient" model is instructed with the patient profile. Both the Single Model and Double Model use the same "Patient" and "Assistant" prompts. Our prompts were built upon existing templates used in previous studies.6

Introduction

Clinical care surveys show that documentation takes up nearly half of clinical time1, correlating with physician burnout and dissatisfaction.2 Large Language Models (LLMs) may reduce this clerical burden by conducting interviews with patients to collect and summarize information prior to the appointment.

Training data is a key part of adjusting LLMs' behavior, in a process known as "fine-tuning".3 This is vital for LLMs in a chatbot role, to prepare the model for handling sensitive information. There is a data scarcity problem in clinical settings due to privacy concerns.4 Synthetic data generation has alleviated some of this demand. However, creating and evaluating synthetic interview transcripts has not been explored.

We created a pipeline using GPT-4 (Fig 1) for automatically generating synthetic semi-structured interview transcripts, with attention to edge case scenarios and patient variations. Code for this pipeline is available on our GitHub (https://github.com/ubcbraincircuits/transcript_generation).

Methods

Patient Prompt

We create synthetic patient with randomized names, ages, occupations, psychiatric concerns, and health histories. We assign either a "rambling" or "brief" typing style and instruct patients to misspell a medication name as an edge case scenario.

Assistant Prompt

Interviews are structured around a manually-created "Question Bank". The Question Bank provides aspects that Assistants must gather from the patient. We instruct the assistant to respond with empathy, clarify misspellings and follow up on incomplete patient responses.

Double/Single Model Transcript Generation

We interact with GPT-4 models by sending different system instructions as a part of a message list (Fig 2).

Transcript Analysis

We use GPT-4 as an evaluator5 to score the transcript based on coverage, edge case adherence, and sentence categorization. We look at characteristics of the patients' messages to evaluate adherence to the given prompt.

Results

Transcripts

We generated transcripts from n=10 synthetic patients using both DM and SM methods, resulting in n=20 transcripts.

Functionality

DM patients with the "rambling" typing style had significantly higher average message lengths (Fig 3a) and lower Distinct-1 scores (Fig 3b). compared with SM patients There were significant differences between the SM and DM transcripts' coverage scores for both roles (Fig 3c). The double model method appears to be more consistent in its coverage even across longer transcripts with larger message list sizes.

Edge Case Scenario

The scenario of the patient misspelling the medication name was carried out by all patients. While most assistants responded appropriately, a minority did not correct the patient as instructed (Fig 3d).

Assistant Behavior

Assistant messages were mostly comprised of empathy statements in all the transcript, with in-depth questions second (Fig 3e).

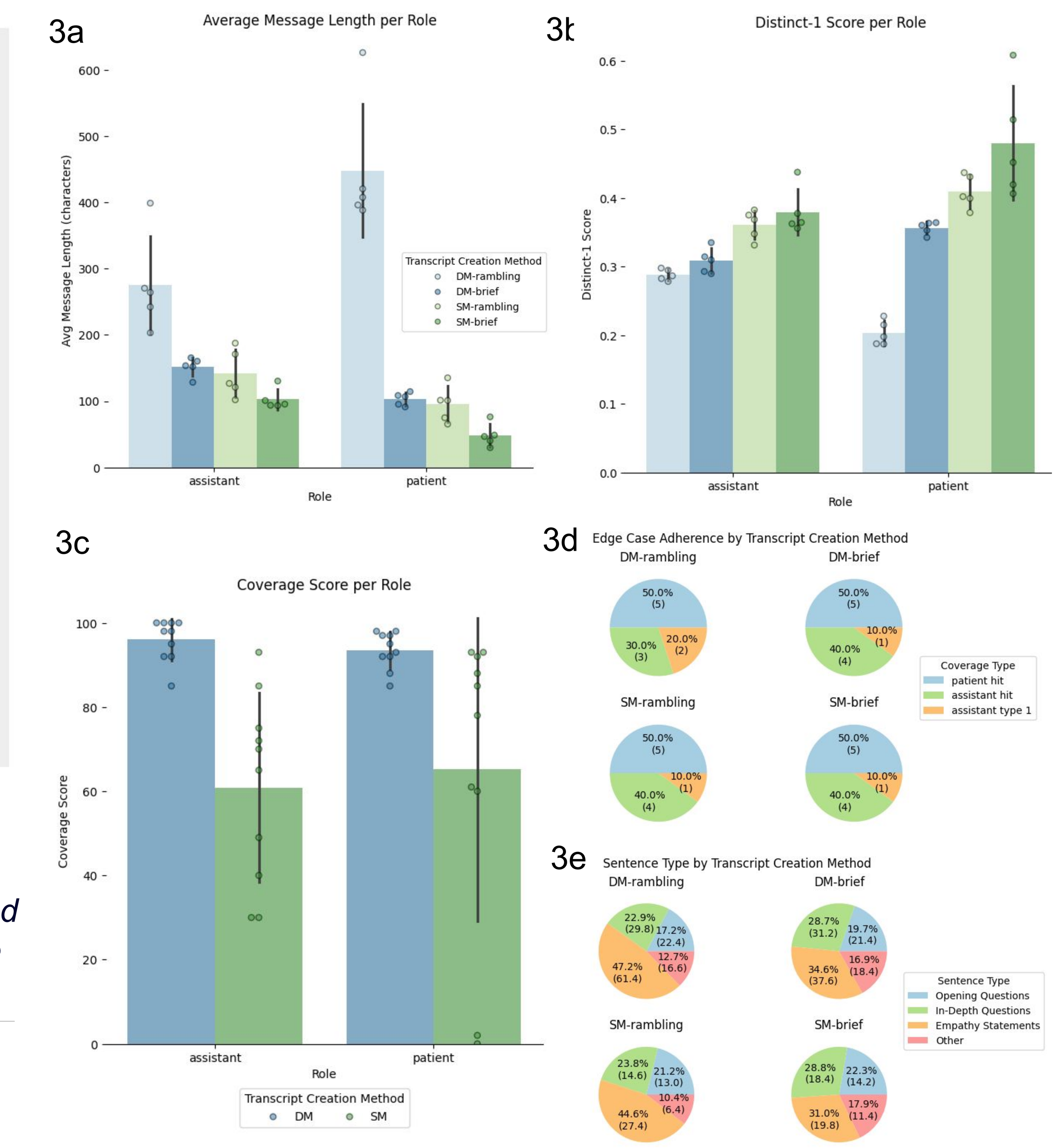

Figure 3a-e Transcript Analysis

Figures are generated by extracting each role's messages to analyze independently. Figures 3c, 3d, 3e are elements obtained using GPT-4.

Conclusion

These statistics suggest that there are noticeable differences in patient performances depending on the generation model used. Our transcript generation pipeline can be a viable method to create a variety of synthetic interview transcripts with distinct patients. This allows for fine-tuning LLMs and further optimizing the quality of synthetically generated datasets in privacy-important settings.

References

1. Sinsky et al., 2016
2. Shanafelt et al., 2016
3. Han et al., 2020
4. Ive, 2022
5. find the source
6. Chen et al., 2022

Acknowledgements